



How smart are crows?

【TED科普：乌鸦到底有多聪明？】

Summary: Recent research reveals that corvids, the family of birds including crows, ravens, and magpies, possess remarkable cognitive abilities. These birds demonstrate strategic food caching, tool use, facial recognition, and even behaviors suggestive of planning for the future and understanding risk. Their intelligence is now considered comparable to that of primates, despite having a very different brain structure. This sophisticated cognition is thought to be crucial for their complex social lives and has been supported by a long developmental period that includes play.

摘要：最新研究表明，包括乌鸦、渡鸦和喜鹊在内的鸦科动物拥有非凡的认知能力。这些鸟类表现出战略性的储食行为、工具使用、面部识别，甚至显示出为未来规划和理解风险的行为。尽管大脑结构截然不同，它们的智力现在被认为可与灵长类动物相媲美。这种复杂的认知能力被认为对其复杂的社会生活至关重要，并得到了包含玩耍在内的较长发育期的支持。





🕒 Estimated Reading Time: 6 min.

📖 四级生词 📖 六级生词 📖 雅思生词 📖 托福生词 📖 专八生词 📖 SAT生词 📖 考研生词 📖
GRE生词

bilibili 00:00:06

🎧 In one of Aesop's 2,500-year-old fables, a crow is searching for water.

在伊索一则2500年前的寓言中，一只#乌鸦#正在寻找水源。

🎧 It spies a pitcher and discovers liquid inside— but it's beyond reach.

它发现了一个#水罐#，看到里面有水——但水位太低够不着。

🎧 Soon, the crow begins dropping pebbles into the pitcher.

很快，#乌鸦#开始往#水罐#里投小石子。

▶ One by one, they displace the water, and the crow quenches its thirst.

石子逐渐#排开#水面，#乌鸦#终于解了#渴#。

▶ Interestingly, this fable turns out to be pretty accurate.

有趣的是，这个#寓言#故事竟相当写实。

bilibili 00:00:30

▶ In a 2014 experiment, scientists placed a food reward in a tube partially filled with water and watched as crow after crow dropped stone after stone in until the food was in beak-snatching distance.

2014年的实验中，科学家将食物放在部分注水的管子里，观察到乌鸦们接连投石入管，直到食物浮至可啄取的高度。

▶ This is just one of many fascinating displays of intelligence from corvids— the bold, brainy family of songbirds that includes crows, ravens, jays, and magpies.

这只是鸦科动物（包括乌鸦、渡鸦、松鸦和喜鹊等大胆聪慧的鸣禽）众多智慧表现中的一例。

▶ To start, corvids are strategic when it comes to their food.

首先，鸦科动物在觅食方面极具#策略性#。

bilibili 00:00:58

▶ They ransack trash bins and snatch bites from under other birds' beaks.

它们会翻捣#垃圾桶#，从其他鸟喙下#抢夺#食物。

▶ They also hoard and cache their food.

它们还懂得#储藏#和#埋藏#食物。

▶ A single Clark's nutcracker, for instance, is thought to scatter some 30,000 pine nuts across more than 2,000 cache sites, which it can recover more than nine months later.

例如，一只克拉克星鸦能将约3万颗松子分散埋藏在2000多个地点，并在九个月后准确找回。

▶ And in one experiment, scrub jays given waxworms and peanuts cached and retrieved the perishable worms first and left the peanuts for later.

另一实验中，获得蜡虫和花生的#灌丛#鸦会优先埋藏并取食#易腐坏#的蠕虫，将花生留待日后。

 00:01:25

⏮ It seemed the jays not only recalled where they buried things, but also what they buried and when.

看来它们不仅记得埋藏地点，还能分辨物品种类与埋藏时间。

⏮ What's more, experiments show that scrub jays will bury food when they know it won't be available the next day, suggesting they can plan for future events too.

更有实验表明，灌丛鸦在预知次日无食时会提前埋粮，这暗示它们具备规划未来的能力。

⏮ And corvid abilities extend beyond meal prepping.

鸦科动物的能力远不止于备餐。

⏮ In one experiment, a masked person trapped and released a wild crow.

某实验中，一名戴#面具#者诱捕并放生了一只野生#乌鸦#。

 00:01:49

⏮ When the masked person returned, crows dive-bombed them.

当该#面具#人再次出现时，乌鸦们以#俯冲#攻击驱赶。

⏮ If they wore a different mask, however, they were ignored, suggesting that corvids can distinguish between— and remember— different human facial characteristics.

但若对方换戴不同#面具#，鸦群便无动于衷，这表明它们能辨别并记忆人类的#面部#特征。

⏮ Crows also tend to gather around the bodies of their dead companions and caw— a behavior sometimes likened to a funeral.

乌鸦常聚集在同伴尸体周围鸣叫——这一行为被比作葬礼。

 00:02:12

⏮ And one study showed that crows were slower to return to an area where they'd seen a dead companion, suggesting they associated it with risk.

研究显示，乌鸦重返见过同伴尸体的区域时会更迟疑，说明它们能将此与危险关联。

▶ Corvids like New Caledonian crows also demonstrate an impressive proclivity for reasoning and tool use.

新喀鸦等鸦科动物还展现出卓越的推理与工具使用天赋。

▶ Even in the wild, they fashion sticks into hook-shaped probes for food.

即使在野外，它们也能将树枝制成钩状探针取食。

▶ Some Japanese crows have been observed placing nuts on the road, waiting nearby, then retrieving the rewards once passing cars have cracked them open.

观察发现，某些日本乌鸦会把坚果放在路上，守候一旁，待车辆碾碎后取食。

 00:02:45

▶ But corvids also do things that are harder to explain.

但鸦科动物也有些难以解释的行为。

▶ They sometimes tote trinkets around, which might help them practice caching or serve as gifts to their partners.

它们有时会携带小物件，可能用于练习储藏，或作为给伴侣的礼物。

▶ And they've been spotted repeatedly sledding down roofs on plastic lids.

还有人观察到它们用塑料盖在屋顶反复#滑行#。

▶ Play activities like this are relatively common in corvids, especially juveniles.

此类游戏行为在鸦科动物中较常见，尤以幼鸟为甚。

 00:03:05

▶ Corvids have a longer developmental period than other songbirds.

鸦科动物的#发育#期比其他鸣禽更长。

▶ And it's thought that play encourages learning and lays a strong foundation for their intelligent behavior.

研究者认为游戏促进学习，为其智能行为奠定基础。

🎧 Some researchers think cleverness is crucial to the highly social lives of corvids.

有学者认为智慧对鸦科高度社会化的生活至关重要。

🎧 They breed cooperatively; live and communicate in large dynamic groups; share food; and mob predators together.

它们合作育雏，在庞大#流动#的群体中生活交流，分享食物，并集体#围攻#天敌。

 00:03:28

🎧 Some have lifelong partnerships where they recognize their companions, and track, respond to, and even predict each other's behaviors and desires.

某些种类会结成终身伴侣，能识别同伴，追踪、响应甚至预判彼此行为与需求。

🎧 Scientists today consider corvids among the smartest animals— a big step from where consensus used to be.

如今科学家将鸦科列为最聪明动物之一——这与昔日#共识#大相径庭。

🎧 In the late 1800s, German scientist Ludwig Edinger compared primate and crow brains and interpreted the crow's brain structure as primitive, dismissing their cognitive abilities.

19世纪末，德国科学家路德维希·埃丁格对比灵长类与乌鸦大脑，认为乌鸦脑结构原始，否定其认知能力。

 00:03:55

🎧 This was a misinterpretation.

这一解读实属谬误。

🎧 A raven, for example, has a brain that looks quite different than a Capuchin monkeys—and is about a third of the size.

例如渡鸦大脑与卷尾猴迥异，体积仅其三分之一。

🎧 But both animals have a similar number of neurons in their pallium, a brain region that's associated with cognition.

但两者大脑皮层（与#认知#相关的脑区）神经元数量相近。

🎵 So while corvid brains travelled a very different evolutionary path than primate brains, it was still one that brought them sophisticated cognition.

因此尽管鸦科与#灵长类#大脑的#进化#路径迥异，却殊途同归地发展出#复杂#的#认知#能力。

 00:04:20

🎵 The next time you spot a crow, you can be assured of its smarts.

下次遇见#乌鸦#时，你尽可确信它的智慧。

🎵 Just maybe don't give it a bad reason to remember you.

只是别给它们留下记恨你的理由。

